

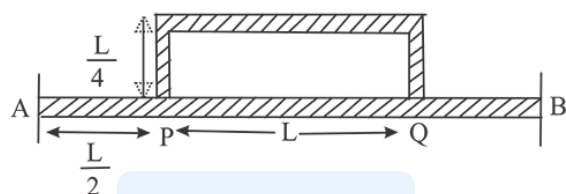
JEE Mains 2019 Chapter wise Question Bank

Heat Transfer - Questions

Q1

Temperature difference of 120°C is maintained between two ends of a uniform rod AB of length $2L$. Another bent

rod PQ, of same cross-section as AB and length $\frac{3L}{2}$, is connected across AB (See figure). In steady state, temperature difference between P and Q will be close to:



- (1) 45°C (2) 75°C
(3) 60°C (4) 35°C

9 Jan Morning

Q2

A heat source at $T = 10^3 \text{ K}$ is connected to another heat reservoir at $T = 10^2 \text{ K}$ by a copper slab which is 1 m thick. Given that the thermal conductivity of copper is $0.1 \text{ WK}^{-1} \text{ m}^{-1}$, the energy flux through it in the steady state is:

- (1) 90 Wm^{-2} (2) 120 Wm^{-2}
(3) 65 Wm^{-2} (4) 200 Wm^{-2}

10 Jan Morning

Q3

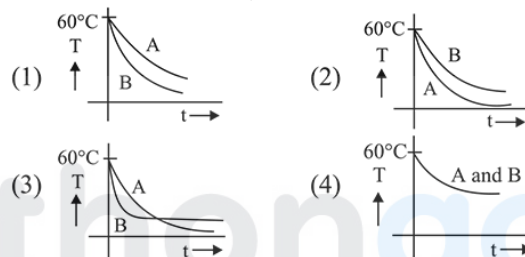
A cylinder of radius R is surrounded by a cylindrical shell of inner radius R and outer radius $2R$. The thermal conductivity of the material of the inner cylinder is K_1 and that of the outer cylinder is K_2 . Assuming no loss of heat, the effective thermal conductivity of the system for heat flowing along the length of the cylinder is:

- (1) $\frac{K_1 + K_2}{2}$ (2) $K_1 + K_2$
(3) $\frac{2K_1 + 3K_2}{5}$ (4) $\frac{K_1 + 3K_2}{4}$

12 Jan Morning

Q4

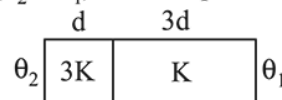
Two identical beakers A and B contain equal volumes of two different liquids at 60°C each and left to cool down. Liquid in A has density of $8 \times 10^2 \text{ kg/m}^3$ and specific heat of $2000 \text{ J kg}^{-1} \text{ K}^{-1}$ while liquid in B has density of 10^3 kg m^{-3} and specific heat of $4000 \text{ J kg}^{-1} \text{ K}^{-1}$. Which of the following best describes their temperature versus time graph schematically? (assume the emissivity of both the beakers to be the same)



8 April Morning

Q5

Two materials having coefficients of thermal conductivity ' $3K$ ' and ' K ' and thickness ' d ' and ' $3d$ ', respectively, are joined to form a slab as shown in the figure. The temperatures of the outer surfaces are ' θ_2 ' and ' θ_1 ', respectively, ($\theta_2 > \theta_1$). The temperature at the interface is:



- (1) $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$ (2) $\frac{\theta_2 + \theta_1}{2}$
(3) $\frac{\theta_1}{6} + \frac{5\theta_2}{6}$ (4) $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$

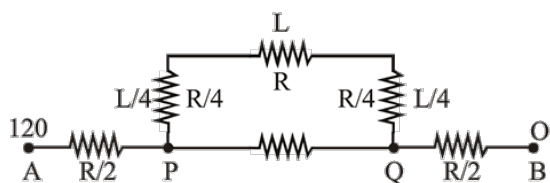
9 April Evening

JEE Mains 2019 Chapter wise Question Bank

Heat Transfer - Answers

Q1

$$(1) \frac{\Delta T_{AB}}{R_{AB}} = \frac{120}{\frac{8}{5}R} = \frac{120 \times 5}{8R}$$

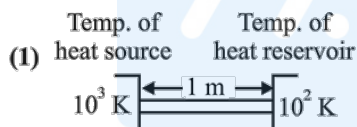


In steady state temperature difference between P and Q,

$$\Delta T_{PQ} = \frac{120 \times 5}{8R} \times \frac{3}{5}R = \frac{360}{8} = 45^\circ \text{C}$$

9 Jan Morning

Q2



$$\left(\frac{dQ}{dt}\right) = \frac{kA\Delta T}{l}$$

$$\text{Energy flux, } \frac{1}{A} \left(\frac{dQ}{dt}\right) = \frac{k\Delta T}{l}$$

$$= \frac{(0.1)(900)}{1} = 90 \text{ W/m}^2$$

10 Jan Morning

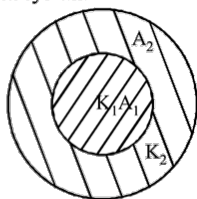
Q3

(4) Effective thermal conductivity of system

$$K_{eq} = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2}$$

$$= \frac{K_1 \pi R^2 + K_2 [\pi (2R)^2 - \pi R^2]}{\pi (2R)^2}$$

$$= \frac{K_1 (\pi R^2) + K_2 (3\pi R^2)}{4\pi R^2} = \frac{K_1 + 3K_2}{4}$$



12 Jan Morning

Q4

$$(2) \text{Rate of Heat loss} = mS \left(\frac{dT}{dt}\right) = e\sigma AT^4$$

$$-\frac{dT}{dt} = \frac{e\sigma \times A \times T^4}{\rho \times Vol. \times S} \Rightarrow -\frac{dT}{dt} \propto \frac{1}{\rho S}$$

$$\left(\frac{dT}{dt}\right)_A = \frac{\rho_B}{\rho_A} \times \frac{S_B}{S_A} = \frac{10^3}{8 \times 10^2} \times \frac{4000}{2000}$$

$$= 2.5 > 1$$

$$\Rightarrow \left(-\frac{dT}{dt}\right)_A > \left(-\frac{dT}{dt}\right)_B$$

So, A cools down at faster rate.

8 April Morning

Q5

$$(1) H_1 = H_2 \theta_2 \left[\frac{d}{3k} \theta_1 \right]$$

$$\text{or } (3k)A \left(\frac{\theta_2 - \theta}{d}\right) = kA \left(\frac{\theta - \theta_1}{3d}\right)$$

$$\text{or } \theta = \left(\frac{\theta_1 + 9\theta_2}{10}\right)$$

9 April Evening